

TW3Cast: A Frozen Router of Lightly Fine-Tuned Foundation Models for Time-Series Forecasting on GIFT-Eval, Selected Entirely on the Training Split*

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Abstract

TW3Cast is a time-series forecasting system that reaches position 3 of 130 entries on the GIFT-Eval benchmark by mean MASE rank, as of 2026-09-14. The two entries above it belong to the leaderboard’s agentic category, multi-step systems that use agents or language models to reason about, generate or select forecasts, and both run a large language model. TW3Cast runs none. Its selection is a table computed once on the training split and then frozen, its experts are public foundation models lightly fine-tuned on those training splits. For each of the 97 dataset, frequency and horizon configurations, the table serves one of four modes: a specialist, which is a LoRA or full fine-tune of Chronos-2, TiRex or Toto whose training data was cleaned and enriched by explicit rules; a quantile blend that contains a specialist; a blend of base models; or a selection tournament played on a backtest carved from the training split. Every decision in the table was taken on that backtest. A specialist is admitted the moment it beats the tournament there, so a candidate costs a few megabytes and minutes of GPU time, and a failed candidate changes nothing. Three guarded mechanisms protect the selection from its own biases: a dual accuracy and calibration criterion, an asymmetric margin against candidates that saw the series during training, and conservative per-window gates. The selection rules themselves were chosen inside a temporal meta-backtest. The best base model served alone reaches a mean MASE rank of 33.8, the tournament served on every configuration reaches 38.0, and the full router reaches 19.4. The routing table, the expert index, the pinned base-model revisions, the submitted score file and the dated snapshot of the public scores are released, and every number in this paper regenerates from them by one script.

1 Introduction

A practitioner who needs one forecast per data source now chooses among several public foundation models, several sizes and several context policies, with fine-tuning as a further option. On the GIFT-Eval benchmark (Aksu et al., 2024), which aggregates 97 configurations over 24 datasets, none of these models served alone reaches the top of the leaderboard. The information that decides which model should speak on which data is in the recent history of each source, and this paper builds a system around that observation.

*Every table, figure and leaderboard number of this paper regenerates by script from the released routing table, the released expert index and a dated snapshot of the public per-configuration scores; see Section 12.

TW3Cast is a router. For every configuration, identified by name, a frozen table gives mixture weights over experts, and the forecast is the weighted, sorted quantile average of the selected experts. The table is not learned end to end. Each row is the outcome of an explicit decision procedure run on a backtest carved out of the training split, and the row is then frozen. At prediction time nothing is decided any more, apart from two per-window guard rails with fixed thresholds. The router can therefore be audited row by row, improved row by row, and shipped as two small files.

Four contributions organize the paper. First, a decision protocol in which every choice is taken on a train-side backtest with a recency bias, and in which the selection rules themselves are validated by a temporal meta-backtest (Section 5). Second, an expert-construction methodology in which light fine-tuning becomes competitive once the data is prepared by explicit rules for per-series cleaning, dosed fusion of sibling datasets, window sampling with robust residual normalization, and a pinball loss on the nine evaluated quantiles (Section 6). Third, a set of guarded selection mechanisms, a tournament with a dual accuracy and calibration criterion and an asymmetric anti-memorization margin, conservative per-window gates, and a correlation guard against misaligned imports (Section 7). Fourth, an accretion property. Admission requires beating the tournament on the backtest, so adding an expert never degrades the system on its selection criterion, and a candidate costs minutes. We report the wins and the configurations where every trained candidate lost and was discarded (Sections 9 and 10). Inserted among the 129 public entries with complete results on 2026-09-14, the system reaches position 3 of 130 by mean MASE rank (Section 9).

2 Related work

Time-series foundation models. The expert pool builds on public pretrained forecasters from three architecture families. Chronos-2 (Ansari et al., 2025) is a multivariate sequence model with joint channel prediction, descended from Chronos (Ansari et al., 2024). TiRex (Auer et al., 2025) is an xLSTM-based univariate model, with a second generation released as a model card (NX-AI, 2026). Toto (Cohen et al., 2025) is a decoder model for observability data, of which we adapt the 2.5B fine-tuned release (Datadog, 2026) with LoRA (Hu et al., 2021). TimesFM (Das et al., 2024; Google Research, 2025) and a public LoRA of Chronos-2 (Vermilion, 2026) complete the tournament pool. Other foundation forecasters, such as Moirai (Woo et al., 2024), Lag-Llama (Rasul et al., 2023) and the mixture-of-experts variants Moirai-MoE (Liu et al., 2024) and Time-MoE (Shi et al., 2025), are not in the pool but appear on the same leaderboard.

Agentic entries. The leaderboard’s agentic category covers multi-step systems that use agents or language models to reason about, generate or select forecasts. The two entries above ours in Section 9 are documented. STRIDE (Ahamed et al., 2026) trains a time-series foundation model on reasoning traces produced by a language model, and Synapse (Das et al., 2025) arbitrates between several foundation models with weights computed at inference. EXAONE-Forecast-Agent (LG AI Research, 2026) wraps a forecasting model in a multi-agent pipeline driven by a large language model. The lower agentic entries have no public description we could read. TW3Cast shares with these systems the idea of routing among public foundation models per data source, and differs in the three choices above: no language model anywhere, a routing decided once on the training split and frozen, and experts obtained by light fine-tuning.

Forecast combination. That combining forecasts beats committing to one goes back to [Bates and Granger \(1969\)](#), and the forecasting competitions have shown simple combinations to be strong ([Makridakis et al., 2020, 2022](#); [Timmermann, 2006](#)). The forecast combination puzzle, in which estimated weights lose to equal weights, has a simple explanation in the estimation noise of the weights ([Smith and Wallis, 2009](#); [Claeskens et al., 2016](#)). Our router uses equal weights inside every blend for that reason, and spends its estimation budget on the choice of members instead.

Model selection by meta-learning. Selecting or weighting forecasters per series from features of the series is the FFORMS and FFORMA line ([Talagala et al., 2023](#); [Montero-Manso et al., 2020](#)), which itself extends earlier meta-learning for forecast selection ([Lemke and Gabrys, 2010](#)). Those methods learn a selector across many series. We select per configuration on a backtest of the configuration’s own history, with guard rails against the biases such a selection creates, in particular the structural advantage of candidates that saw the series during training. Mixture-of-experts routing ([Shazeer et al., 2017](#)) learns the router jointly with the experts. We compute the routing table by measurement and freeze it, which trades expressiveness for auditability and zero inference-time overhead.

Backtesting for selection. The choice between blocked temporal splits and random splits for time-series model selection has a long literature ([Tashman, 2000](#); [Bergmeir and Benítez, 2012](#); [Cerqueira et al., 2020](#)). Our temporal meta-backtest applies the same choice one level up, to the selection rule rather than to the model.

3 Problem statement

Definition 1 (Configuration). *A configuration is a triple of dataset, sampling frequency and horizon class in the benchmark. Each configuration provides a training split and a sequence of official evaluation windows. GIFT-Eval has 97 configurations. Each per-window forecast is scored by MASE ([Hyndman and Koehler, 2006](#)) for point accuracy and by the mean weighted quantile loss over the nine deciles for calibration, a discrete approximation of the CRPS ([Gneiting and Raftery, 2007](#)). Leaderboard positions aggregate per-configuration ranks.*

Problem 1 (Per-configuration serving under zero test access). *For each configuration, choose the forecaster to serve, a single model, a quantile blend or a per-window policy over a pool, using only the training split, so that the mean per-configuration rank over all 97 configurations is minimized.*

Definition 2 (Admission rule). *Let $S(\cdot)$ denote the backtest score of a candidate under the dual criterion of Section 7. A trained specialist is designated for a configuration as soon as it beats the tournament’s champion on the backtest. Under comparable scores the specialist is preferred. Otherwise the configuration keeps the tournament as its serving mode.*

The admission rule makes improvement monotone by construction on the selection criterion, and it prices a failed experiment at the cost of training the candidate.

4 The router

The shipped system is a routing table `router.csv` with one row per configuration and expert, giving a mixture weight, and an expert index `experts.json` giving each expert’s architecture

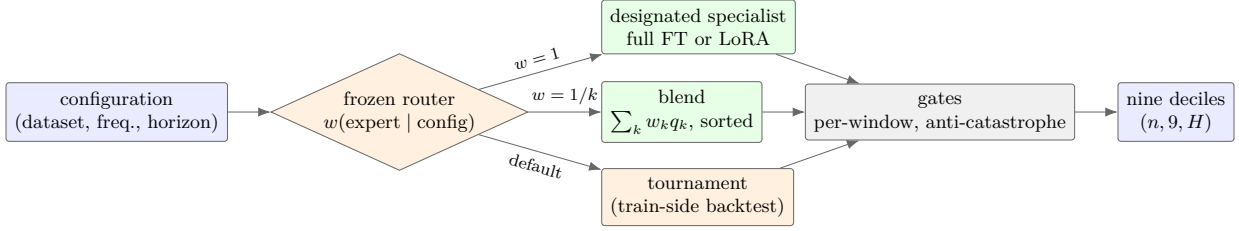


Figure 1: The frozen router. A configuration name resolves by table lookup to a designated specialist, a quantile blend or the tournament mode. Per-window gates then bound the tail risk, and the system outputs the nine deciles for each evaluation window.

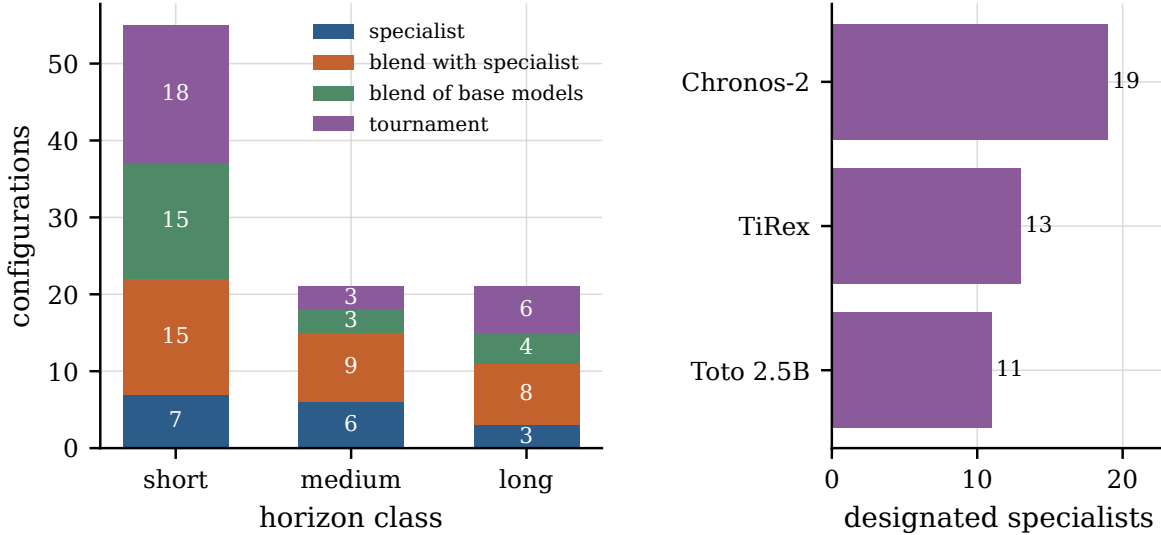


Figure 2: Composition of the released router. Left, the serving mode of the 97 configurations by horizon class. Right, the architecture family of the 43 designated specialists. Both panels are read from `router.csv` and `experts.json` by `make_figures.py`.

family. Weights in a row sum to one. Figure 1 shows the four serving modes that the table encodes. A designated specialist has weight one. A blend has equal weights $1/k$ over its k members, which may mix specialists and base models. The reserved member `__tournament__` marks a configuration whose forecast is produced by the train-side tournament of Section 7. Inference is a table lookup, the experts’ forward passes and a sorted weighted average of quantiles.

In the released table, 16 configurations are served by a single specialist, 32 by a blend that contains at least one specialist, 22 by a blend of base models only, and 27 by the tournament. The 43 designated specialists split into 19 Chronos-2 experts, 13 TiRex experts and 11 Toto experts. Blends have two members in 23 configurations, three in 30 and four in 1. Figure 2 gives the breakdown by horizon class, and Appendix B lists the full table.

5 The decision protocol

A backtest carved from the training split. All decisions are taken on context-plus-horizon windows sliced from the end of the training split. Recent windows resemble the conditions the

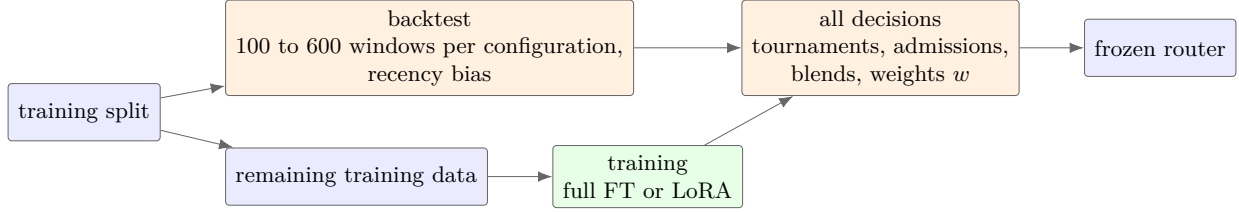


Figure 3: Every decision is taken on a backtest extracted from the training split. The remaining training data trains the experts, and the resulting router is frozen before any test window is read.

forecaster will face, and on long series old regimes mislead selection. Windows are cut at several positions per series, hourly regimes are identified, and each configuration receives roughly 100 to 600 backtest windows. Candidates are scored there as the benchmark scores them, with MASE computed on the real context before any padding, and with the CRPS approximated by the mean pinball loss $\rho_q(y, \hat{y}_q) = \max(q(y - \hat{y}_q), (q - 1)(y - \hat{y}_q))$ over $q \in \{0.1, \dots, 0.9\}$ (Koenker and Bassett, 1978). The training data not consumed by the backtest trains the specialists (Figure 3).

Validating the selector itself. Before trusting a selection rule we evaluate it by a temporal meta-backtest. The rule selects on the older backtest windows and its choice is scored on the recent ones, over hundreds of draws. The split is temporal, following Bergmeir and Benítez (2012) and Cerqueira et al. (2020) one level up, from models to selection rules. All rule iteration in this work, on margins, tie-breaks and gate thresholds, was carried out inside this protocol, at no cost in evaluation data.

6 Building the experts

The design principle of this section is that the value of a fine-tune is decided before training starts, by how its data is prepared. Each preparation rule below is cheap and explicit, and every candidate it produces faces the admission rule of Definition 2.

Cleaning. Every corpus is filtered series by series before training. Degenerate series with near-zero variance, constant or few-valued, crush the loss while teaching nothing. Scale outliers from failing sensors or mixed units can dominate the gradient on their own. Corrupted stretches are truncated instead of imputed. Per-source volume caps in fusions stop any one dataset from drowning the others. Cleaning is contextual. On a 21-series weather dataset, fusing with a large neighboring corpus drowned the signal and the cleaned solo won, and the opposite held for data-poor datasets. The rule that emerged is that a data-rich configuration wants a dedicated specialist on cleaned data, while a data-poor configuration wants an enriched family.

Dosed fusion. For data-poor configurations the target’s training data is mixed with that of sibling configurations of the same domain or frequency, with the target over-weighted, typically three parts target to one part siblings. The siblings regularize and the target remains the signal. Families are built by similarity, first frequency and horizon, then domain, and entirely poor families borrow from rich siblings. Rich families receive full fine-tuning and poor ones LoRA. Unstable targets get several seeds, and a short run of 500 steps is extended to 2,000 only if it already wins.

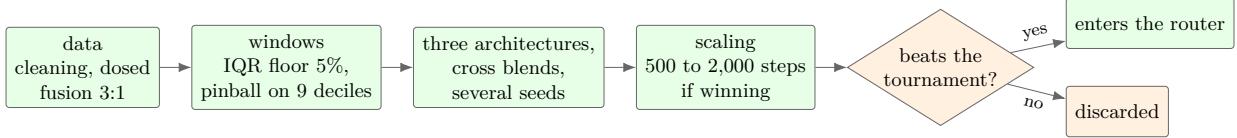


Figure 4: The expert pipeline. Cleaning and dosed fusion, controlled windows with robust normalization and pinball loss, three architectures with cross blends and conditional scaling, then the admission rule.

Controlled windows. Training reproduces the evaluation conditions. Windows are sampled with variable-length contexts, the target is the full horizon or the next patch depending on the architecture, and residuals are normalized robustly,

$$z = \frac{x - \text{med}(c)}{s(c)}, \quad s(c) = \max(\text{IQR}(c), 0.05 \cdot \text{range}(c)), \quad |z| \leq 20,$$

without which the loss is dominated by outliers. The loss is the pinball loss on the benchmark’s own nine deciles, with sorted outputs, so the model learns the distribution it will be judged on. Learning rates are 10^{-4} for LoRA and 10^{-5} for full fine-tunes.

Three architectures and cross blends. Chronos-2, TiRex and Toto 2.5B are specialized, each with its native recipe. Blends are sorted quantile means across architectures, so that the members of a blend fail on different windows. Each candidate is a checkpoint of a few megabytes trained in minutes, and by Definition 2 it enters the router only by beating the tournament on the backtest (Figure 4).

7 The tournament and its guard rails

Stage 1, a selection defended against its own biases. On configurations without a designated specialist, every pool member is scored on the backtest windows and the winner becomes the served default. Three rules make this selection robust. A dual criterion requires both MASE and CRPS, because members with degenerate quantiles, nine identical deciles, can win MASE alone while being probabilistically useless. An asymmetric anti-memorization margin handles locally trained candidates, whose backtest scores are flattered because they saw these series during training. Such a candidate is admitted only if $S(\text{local}) < (1 - \delta) \cdot \min_{\text{generic}} S$ with $\delta = 12\%$, and only if it is not worse on CRPS. Near-ties, with a gap below 3%, are settled on CRPS. When the blend of the pool’s two largest architectures is within 5% of the winner, the blend is served instead, because a win by a hair is usually backtest noise and blending is the safer choice under uncertainty.

Stage 2, a conservative per-window gate. At prediction time each window is checked by a lightweight backtest of its own context on two segments, the final segment of horizon length and the middle segment. A window is rerouted only if all conditions agree. The default’s error must exceed a threshold, $\tau = 1.5$ for foundation models and $\tau = 3$ for naive references. The challenger must be better by at least 10% on average, better on each segment, and better on CRPS. A regime-transition guard blocks any switch when the recent dormancy of the context differs from what the segments saw, as at dawn and dusk of hourly series, where the segment signal inverts. The gate is designed to switch rarely.

Safety nets for designated experts. Configurations served by a specialist carry a distinct anti-catastrophe gate with high thresholds, $\tau = 5$ to 10. It intervenes only on flagrant per-window failure, in which case the window falls back to the generic pool, and it never trims a healthy specialist. An alignment guard verifies by correlation that imported predictions from caches or external wrappers are attached to the right windows. Plausible-looking but misaligned outputs are the worst kind of error, since aggregate metrics hide them, and they are rejected when the median correlation falls below 0.30. Over our campaigns this guard fired three times, each time on a real misalignment, and never on an aligned import.

8 Evaluation protocol

Harness. All reported scores come from the official GIFT-Eval harness with the leaderboard parameters, which is the forecast evaluation routine of gluonts (Alexandrov et al., 2020). We validated our installation by reproducing a published reference model to three decimals. Every submitted line is recomposed from a saved artifact, weights, adapters or verified quantiles, and checked for equality on MASE and weighted quantile loss before submission.

Leaderboard snapshot. The public leaderboard publishes one per-configuration result file per entry. On 2026-09-14 we downloaded every result file and every entry description from the leaderboard repository. Entries with a result file covering all 97 configurations number 129. For each configuration we rank all entries by MASE and by weighted quantile loss, ties sharing the lowest rank, and we average the ranks over the 97 configurations. Our submission is inserted into this set as one more entry and ranked by the same rule. Every pool member declares no training overlap with the test sets in its leaderboard description, and our own training touches only the provided training splits. The leaderboard is living, so every position in this paper is dated. The snapshot itself is part of the release (Section 12).

9 Results

Result 1 (standing on the leaderboard). Inserted among the 129 public entries, TW3Cast reaches position 3 of 130 by mean MASE rank, with a mean rank of 19.4. Every entry above it is declared as agentic by its authors, the category described in Section 2, so TW3Cast is the best entry that runs no agent and no language model. By mean weighted quantile loss rank the system is at position 8, with a mean rank of 24.6, so its calibration lags its point accuracy. Table 1 gives the top of the leaderboard and Figure 5 the full distribution.

Result 2 (components alone do not explain the position). The building blocks are plain public foundation models, each of which is also a public leaderboard entry, and Table 2 gives their own standing. The best of them, Toto 2.0 2.5B FT, sits at position 23 with a mean MASE rank of 33.8, and the weakest pool member has a mean rank of 58.4. A harder comparison is the per-configuration oracle that picks, on each configuration, the pool member with the lowest test MASE, a choice that requires test knowledge. Against that oracle the system is better on 37 of the 97 configurations, within five percent on 60, and more than ten percent worse on 30 (Figure 6). The gap between the components and the complete system is created by data preparation, training and guarded selection, since nothing else enters the system.

Position	Entry	Mean MASE rank	Declared type	Test leakage
1	STRIDE_w_Synapse	14.9	agentic	No
2	EXAONE-Forecast-Agent	19.2	agentic	No
3	TW3Cast (this work)	19.4	router	No
4	LS-MoE	19.7	agentic	No
5	LS-Agent	21.7	agentic	No
6	Falcon-Agent	22.5	agentic	No
7	CastStar	22.6	agentic	No
8	limix_moe	22.7	agentic	No

Table 1: Top of the GIFT-Eval leaderboard by mean per-configuration MASE rank on 2026-09-14, computed by `make_figures.py` from the per-configuration result files of the 129 public entries with complete results, with the submitted TW3Cast result file inserted as one more entry and ranked by the rule of Section 8. Declared type and leakage are copied from each entry’s own description file on the leaderboard. The snapshot of the public files, our result file and the script are in the release.

Pool member	Position	MASE rank	WQL rank	Declared type
Chronos-2	44	47.2	47.6	pretrained
TiRex	60	58.4	51.1	zero-shot
TiRex-2	36	44.2	42.7	pretrained
Toto 2.0 2.5B FT	23	33.8	33.1	fine-tuned
TimesFM 2.5	57	53.5	53.1	zero-shot
TurkForecast LoRA	45	48.1	47.2	fine-tuned

Table 2: Standing of each tournament pool member as its own public leaderboard entry, in the same field and with the same rank computation as Table 1. The WQL column is the same average on weighted quantile loss. TiRex-2 is listed under its pretrained entry, the one used in the pool.

Result 3 (the tournament and the specialists each earn their place). Table 3 ranks, in the same public field, each pool member served alone, a variant in which the tournament serves every configuration with no designated specialist, and the released router. Served on every configuration the tournament reaches a mean MASE rank of 38.0, behind the best single member at 33.8. On the 27 configurations that the released router actually assigns to the tournament, the same tournament reaches 19.8 where the best member reaches 28.1. On the other 70 configurations the designated specialist or blend beats the tournament-only serving on 69, with a median MASE gain of 2.0 percent. The router is worth more than either of its halves because it assigns each half the configurations it wins.

Result 4 (where training won and where it lost). The admission rule makes negative results cheap and visible. On several configurations, among them two short-horizon business-monitoring datasets and an hourly weather dataset, every fine-tuned candidate we trained across the three architectures lost to the tournament on the backtest and was discarded. On those configurations the tournament’s cross-architecture blend with the per-window gate is the served mode. Each discarded candidate cost minutes of GPU time and changed nothing in the released system. The same mechanism captured the wins. On a 10-minute weather configuration a cleaned-solo LoRA of Chronos-2 beat the tournament and was designated, and on the 48 configurations served by or with a specialist the designated expert beat the tournament on the backtest by construction.

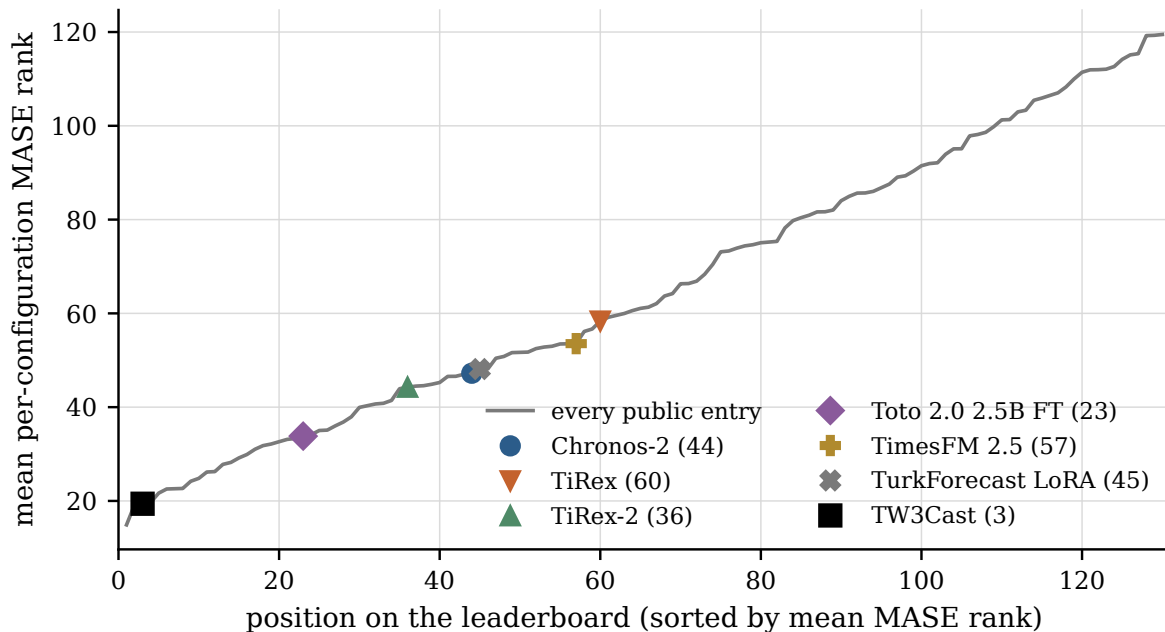


Figure 5: Mean per-configuration MASE rank of the 130 entries of Table 1, sorted by position, with the six members of the tournament pool and TW3Cast marked; the number in parentheses is the position. Script `make_figures.py`.

10 Deployability and accretion

The shipped system is two small files and a set of checkpoints. Inference is a table lookup, the experts’ forward passes and a sorted weighted average of quantiles, with no agent and no language model in the loop.

The router improves by accretion. A candidate expert is a LoRA adapter of a few megabytes, trained in 500 to 2,000 steps, two to seven minutes on a single GPU. Admission requires beating the tournament on the train-side backtest, so adding an expert never degrades the system on its selection criterion, and a failed candidate costs minutes and changes nothing. No global retraining ever occurs. Integrating, replacing or retiring an expert is a one-row change in the table.

The same property drives adaptation beyond the benchmark, because nothing in the method depends on configuration names. On a new data source the tournament builds its backtest on the available history and serves a default with the same guard rails, with no training at all. When specialization is worth trying, the pipeline of Section 6 applies to the source’s history and the admission rule arbitrates. The router grows wherever cheap training wins and falls back to its tournament wherever it does not.

11 Threats to validity

A living leaderboard. Positions move with every new submission. Every rank in this paper is therefore computed from a dated snapshot of the public per-configuration files, which the release contains, and the rank computation is a short script rather than a reading of the leaderboard page. The snapshot fixes the comparison set but does not fix the future, and a later reader should expect a different position.

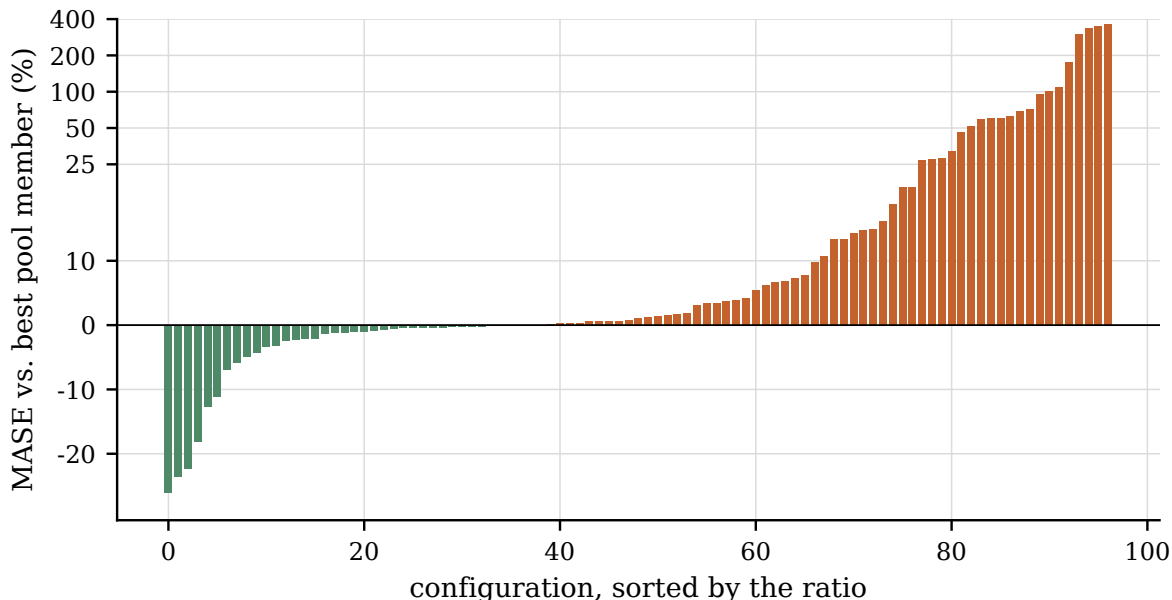


Figure 6: Per-configuration MASE of TW3Cast relative to the best pool member on that configuration, in percent, over the 97 configurations sorted by the ratio. Negative bars are configurations where the system beats the per-configuration oracle over the pool. Script `make_figures.py`.

Recency bias as an assumption. The backtest samples the end of the training split on purpose. This is an assumption about regime continuity, not a theorem. The temporal meta-backtest is built on the same assumption, and a source with an abrupt regime change at the train and test boundary would defeat both.

Selection flexibility. Any system with margins and thresholds can overfit its own validation data. All such constants (Appendix A) were chosen inside the temporal meta-backtest and then frozen, and none was tuned on official evaluation results. The protection is procedural, and it does not remove the dependence of the constants on this benchmark’s mix of frequencies and domains.

Calibration. The system is selected and admitted under a dual MASE and CRPS criterion, yet its weighted quantile loss rank (position 8) lags its MASE rank (position 3). The dual criterion ranks on MASE and uses CRPS as a veto, so a candidate that wins on accuracy with adequate calibration is preferred to one that wins on calibration. A CRPS-first admission rule is the obvious variant and is not evaluated here.

Serialization drift. One serialization path of a base library drifted by under one percent between the in-memory model and its reloaded checkpoint. Only reloaded artifacts may produce published numbers under our rule, one designated expert was re-scored from its reloaded checkpoint for this reason, and checkpoint formats with bit-exact reload are preferred throughout. One expert whose seed-averaged weights were not retained is shipped as its verified forecast quantiles instead, which reproduces its forecasts but not its training.

Variant	All 97	Tournament configurations (27)
Chronos-2	47.2	42.6
TiRex	58.4	63.2
TiRex-2	44.2	41.5
Toto 2.0 2.5B FT	33.8	28.1
TimesFM 2.5	53.5	58.8
TurkForecast LoRA	48.1	51.4
Tournament on every configuration	38.0	19.8
Full system (released router)	19.4	19.7

Table 3: Mean per-configuration MASE rank of each variant in the field of Table 1, over all 97 configurations and over the 27 configurations that the released router serves by tournament. Pool members are their public entries and keep the ranks of Table 2. The tournament-only variant is scored from its released per-configuration grid, inserted into the same field as one more entry; that grid coincides with the submission on the tournament configurations. Script `make_figures.py`.

12 Reproducibility

The code repository¹ contains the routing table `router.csv`, the expert index `experts.json`, the pinned revisions of the six base models `base_models.json`, and a reference inference script `predict.py`. Expert checkpoints are distributed with the repository under their expert identifiers, and the one expert shipped as quantiles is loaded from its quantile file. The paper’s own folder in the repository contains the dated snapshot of the public per-configuration result files, the submitted TW3Cast result file, the per-configuration grid of the tournament-only variant, and the script `make_figures.py` that regenerates every table, figure and number of this paper from them.

Two regimes apply. The exact regime regenerates every table, figure and number of this paper from the release alone, with no model call, in seconds. The fresh regime regenerates our result file by running the inference script on each configuration and scoring it with the official harness, which requires the base models at their pinned revisions, the expert checkpoints and one GPU. The tournament procedure for the 27 configurations it serves is described in Section 7 and reproduces the served forecasts from the pool members and the training split.

13 Conclusion

A frozen table of mixture weights over lightly fine-tuned experts, fed by an explicit data-preparation methodology and defended by guarded train-side selection, reaches position 3 of 130 on GIFT-Eval by mean MASE rank on 2026-09-14 once inserted among the 129 public entries, with no agent and no language model at inference. The design optimizes for properties that large monolithic forecasters lack. Every routing row traces to a measured decision, admission by beating the tournament makes each upgrade individually verified at minutes per candidate, and the same machinery produces a router for any new set of series. The accretion property is the consequence we expect to matter most in practice, because it turns the maintenance of a forecasting system from periodic retraining into a stream of small, cheap, verified changes.

¹<https://github.com/TW3-Partners-OS/TW3-Cast>

References

- Md Atik Ahamed, Mihir Parmar, Palash Goyal, Chun-Liang Li, Qiang Cheng, Tomas Pfister, et al. Reasoning-aware training for time series forecasting. *arXiv preprint arXiv:2605.08625*, 2026.
- Taha Aksu, Gerald Woo, Juncheng Liu, Xu Liu, Chenghao Liu, Silvio Savarese, Caiming Xiong, and Doyen Sahoo. GIFT-Eval: A benchmark for general time series forecasting model evaluation. *arXiv preprint arXiv:2410.10393*, 2024.
- Alexander Alexandrov, Konstantinos Benidis, Michael Bohlke-Schneider, Valentin Flunkert, Jan Gasthaus, Tim Januschowski, Danielle C. Maddix, Syama Rangapuram, David Salinas, Jasper Schulz, et al. GluonTS: Probabilistic and neural time series modeling in Python. *Journal of Machine Learning Research*, 21(116):1–6, 2020.
- Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, et al. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024.
- Abdul Fatir Ansari, Oleksandr Shchur, Jaris Küken, Andreas Auer, Boran Han, Pedro Mercado, Syama Sundar Rangapuram, Huibin Shen, Lorenzo Stella, Xiyuan Zhang, et al. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2025.
- Andreas Auer, Patrick Podest, Daniel Klotz, Sebastian Böck, Günter Klambauer, and Sepp Hochreiter. TiRex: Zero-shot forecasting across long and short horizons with enhanced in-context learning. *arXiv preprint arXiv:2505.23719*, 2025.
- John M. Bates and Clive W. J. Granger. The combination of forecasts. *Journal of the Operational Research Society*, 20(4):451–468, 1969.
- Christoph Bergmeir and José M. Benítez. On the use of cross-validation for time series predictor evaluation. *Information Sciences*, 191:192–213, 2012.
- Vitor Cerqueira, Luís Torgo, and Igor Mozetič. Evaluating time series forecasting models: An empirical study on performance estimation methods. *Machine Learning*, 109:1997–2028, 2020.
- Gerda Claeskens, Jan R. Magnus, Andrey L. Vasnev, and Wendun Wang. The forecast combination puzzle: A simple theoretical explanation. *International Journal of Forecasting*, 32(3): 754–762, 2016.
- Ben Cohen, Emaad Khwaja, Youssef Doubli, Salahidine Lemaachi, Chris Lettieri, Charles Masson, Hugo Miccinilli, Elise Ramé, Qiqi Ren, Afshin Rostamizadeh, et al. This time is different: An observability perspective on time series foundation models. *arXiv preprint arXiv:2505.14766*, 2025.
- Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. In *International Conference on Machine Learning*, 2024.
- Sarkar Snigdha Sarathi Das, Palash Goyal, Mihir Parmar, Yiwen Song, Long T. Le, Lesly Miculicich, et al. Synapse: Adaptive arbitration of complementary expertise in time series foundational models. *arXiv preprint arXiv:2511.05460*, 2025.

- Datadog. Toto 2.0 2.5b-ft (model release). <https://huggingface.co/Datadog/Toto-2.0-2.5B-FT>, 2026.
- Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- Google Research. TimesFM 2.5 200m (model release). <https://huggingface.co/google/timesfm-2.5-200m-pytorch>, 2025.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
- Rob J. Hyndman and Anne B. Koehler. Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4):679–688, 2006.
- Roger Koenker and Gilbert Bassett. Regression quantiles. *Econometrica*, 46(1):33–50, 1978.
- Christiane Lemke and Bogdan Gabrys. Meta-learning for time series forecasting and forecast combination. *Neurocomputing*, 73(10–12):2006–2016, 2010.
- LG AI Research. EXAONE-Forecast (code and model release). <https://github.com/LGAI-Research/EXAONE-Forecast>, 2026.
- Xu Liu, Juncheng Liu, Gerald Woo, Taha Aksu, Yuxuan Liang, Roger Zimmermann, Chenghao Liu, Silvio Savarese, Caiming Xiong, and Doyen Sahoo. Moirai-MoE: Empowering time series foundation models with sparse mixture of experts. *arXiv preprint arXiv:2410.10469*, 2024.
- Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. The M4 competition: 100,000 time series and 61 forecasting methods. *International Journal of Forecasting*, 36(1):54–74, 2020.
- Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting*, 38(4):1346–1364, 2022.
- Pablo Montero-Manso, George Athanasopoulos, Rob J. Hyndman, and Thiyanga S. Talagala. FFORMA: Feature-based forecast model averaging. *International Journal of Forecasting*, 36(1):86–92, 2020.
- NX-AI. TiRex-2 (model release). <https://huggingface.co/NX-AI/TiRex-2>, 2026.
- Kashif Rasul, Arjun Ashok, Andrew Robert Williams, Hena Ghonia, Rishika Bhagwatkar, Arian Khorasani, Mohammad Javad Darvishi Bayazi, George Adamopoulos, Roland Riachi, Nadhir Hassen, et al. Lag-Llama: Towards foundation models for probabilistic time series forecasting. *arXiv preprint arXiv:2310.08278*, 2023.
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations*, 2017.
- Xiaoming Shi, Shiyu Wang, Yuqi Nie, Dianqi Li, Zhou Ye, Qingsong Wen, and Ming Jin. Time-MoE: Billion-scale time series foundation models with mixture of experts. In *International Conference on Learning Representations*, 2025.

- Jeremy Smith and Kenneth F. Wallis. A simple explanation of the forecast combination puzzle. *Oxford Bulletin of Economics and Statistics*, 71(3):331–355, 2009.
- Thiyanga S. Talagala, Rob J. Hyndman, and George Athanasopoulos. Meta-learning how to forecast time series. *Journal of Forecasting*, 42(6):1476–1501, 2023.
- Leonard J. Tashman. Out-of-sample tests of forecasting accuracy: An analysis and review. *International Journal of Forecasting*, 16(4):437–450, 2000.
- Allan Timmermann. Forecast combinations. In *Handbook of Economic Forecasting*, volume 1, pages 135–196. Elsevier, 2006.
- Verm1ion. TurkForecast-FM: A public LoRA adapter of Chronos-2 (model release). <https://huggingface.co/Verm1ion/turkforecast-fm-chronos2-lora-v1>, 2026.
- Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Unified training of universal time series forecasting transformers. In *International Conference on Machine Learning*, 2024.

A Constants and thresholds

Mechanism	Constant	Value	Set by
Anti-memorization margin	δ	12%	meta-backtest
Near-tie CRPS tie-break	gap	< 3%	meta-backtest
Blend fallback	gap to winner	< 5%	meta-backtest
Per-window gate	τ (foundation / naive)	1.5 / 3.0	meta-backtest
Per-window gate	challenger advantage	$\geq 10\%$	meta-backtest
Anti-catastrophe gate	τ	5 to 10	meta-backtest
Alignment guard	median correlation	≥ 0.30	incident analysis
Normalization	scale floor	$0.05 \cdot \text{range}$	training stability
Normalization	residual clip	$ z \leq 20$	training stability
Fine-tuning	LoRA / full learning rate	10^{-4} / 10^{-5}	standard
Fine-tuning	steps (short / extended)	500 / 2,000	conditional scaling
Backtest	windows per configuration	100 to 600	budget

Table 4: All fixed constants of the released system. Every constant was chosen inside the train-side protocols of Sections 5 and 7 and then frozen.

B The released routing table

Tables 5 and 6 list the served mode of every configuration as read from `router.csv`. Abbreviations are C2 for Chronos-2, TFM for TimesFM 2.5, Toto for Toto 2.0 2.5B FT, Toto-uni for a univariate Toto member, Turk for the TurkForecast LoRA of Chronos-2, and TiRex and TiRex-2 for the two TiRex releases.

Configuration	Served by
bitbrains_fast_storage/5T/long	tournament
bitbrains_fast_storage/5T/medium	E01 (Toto)
bitbrains_fast_storage/5T/short	tournament
bitbrains_fast_storage/H/short	E02 (Toto)
bitbrains_rnd/5T/long	Toto + C2 + Turk
bitbrains_rnd/5T/medium	Toto + C2 + Turk
bitbrains_rnd/5T/short	tournament
bitbrains_rnd/H/short	C2 + Turk
bizitobs_application/10S/long	tournament
bizitobs_application/10S/medium	tournament
bizitobs_application/10S/short	tournament
bizitobs_l2c/5T/long	E03 (C2)
bizitobs_l2c/5T/medium	E03 (C2)
bizitobs_l2c/5T/short	tournament
bizitobs_l2c/H/long	E42 (Toto)
bizitobs_l2c/H/medium	E42 (Toto)
bizitobs_l2c/H/short	Toto + C2 + TFM
bizitobs_service/10S/long	tournament
bizitobs_service/10S/medium	Toto + C2 + TFM
bizitobs_service/10S/short	tournament
car_parts/M/short	C2 + TFM + TiRex-2
covid_deaths/D/short	tournament
electricity/15T/long	E09 (Toto) + C2 + TiRex
electricity/15T/medium	E09 (Toto) + C2 + Turk
electricity/15T/short	tournament
electricity/D/short	E04 (C2) + Toto
electricity/H/long	E05 (C2)
electricity/H/medium	E06 (Toto) + C2 + E07 (C2)
electricity/H/short	Toto + C2 + TiRex
electricity/W/short	E08 (C2)
ett1/15T/long	Turk + C2 + E09 (Toto) + TiRex
ett1/15T/medium	E10 (C2) + C2
ett1/15T/short	Toto + C2 + TiRex
ett1/D/short	E11 (C2) + C2
ett1/H/long	TiRex + E12 (TiRex) + TiRex-2
ett1/H/medium	E13 (C2) + Toto-uni
ett1/H/short	tournament
ett1/W/short	TFM + TiRex-2 + C2
ett2/15T/long	E14 (C2) + Turk
ett2/15T/medium	E10 (C2) + E13 (C2)
ett2/15T/short	Toto + C2 + TFM
ett2/D/short	E04 (C2) + E15 (C2)
ett2/H/long	Turk + TFM + E16 (Toto)
ett2/H/medium	E13 (C2) + Toto-uni
ett2/H/short	E17 (TiRex) + C2
ett2/W/short	E18 (Toto)
hierarchical_sales/D/short	Toto + C2 + TiRex
hierarchical_sales/W/short	E19 (C2)
hospital/M/short	tournament

Table 5: The released routing table, first half of the 97 configurations of `router.csv` in alphabetical order, with the members of each blend. Members named *Enn* are designated specialists, with their architecture family in parentheses; other names are base models of the tournament pool at their pinned revisions; blend members carry equal weights. Generated by `make_figures.py`.

Configuration	Served by
jena_weather/10T/long	tournament
jena_weather/10T/medium	E44 (C2)
jena_weather/10T/short	tournament
jena_weather/D/short	E20 (TiRex) + TFM
jena_weather/H/long	tournament
jena_weather/H/medium	E43 (C2) + TFM + C2
jena_weather/H/short	tournament
kdd_cup_2018/D/short	Toto + TFM + Turk
kdd_cup_2018/H/long	E22 (TiRex) + TiRex
kdd_cup_2018/H/medium	E22 (TiRex) + TiRex
kdd_cup_2018/H/short	tournament
loop_seattle/5T/long	tournament
loop_seattle/5T/medium	tournament
loop_seattle/5T/short	Toto + C2 + TFM
loop_seattle/D/short	E23 (TiRex) + TFM
loop_seattle/H/long	Toto + C2 + TFM
loop_seattle/H/medium	E24 (Toto)
loop_seattle/H/short	E27 (Toto) + TFM + C2
m4_daily/D/short	E25 (TiRex) + E26 (TiRex)
m4_hourly/H/short	E27 (Toto) + TiRex + TFM
m4_monthly/M/short	tournament
m4_quarterly/Q/short	E32 (C2) + Toto
m4_weekly/W/short	Toto + TFM
m4_yearly/A/short	tournament
m_dense/D/short	Turk + E28 (Toto) + TFM
m_dense/H/long	Toto + C2 + TFM
m_dense/H/medium	Toto + C2 + TFM
m_dense/H/short	E29 (TiRex) + C2 + E27 (Toto)
restaurant/D/short	E30 (C2) + Turk + E31 (TiRex)
saugeen/D/short	C2 + TFM
saugeen/M/short	Toto + C2 + TFM
saugeen/W/short	Toto + Turk
solar/10T/long	Toto + C2 + Turk
solar/10T/medium	tournament
solar/10T/short	Toto + C2 + Turk
solar/D/short	tournament
solar/H/long	TiRex + E33 (TiRex) + E34 (TiRex)
solar/H/medium	E35 (TiRex)
solar/H/short	tournament
solar/W/short	E19 (C2)
sz_taxi/15T/long	E36 (TiRex) + Toto
sz_taxi/15T/medium	E36 (TiRex) + Toto
sz_taxi/15T/short	tournament
sz_taxi/H/short	E27 (Toto) + E37 (C2)
temperature_rain/D/short	E41 (C2)
us_births/D/short	tournament
us_births/M/short	E38 (C2) + E39 (C2)
us_births/W/short	E40 (Toto)

Table 6: The released routing table, second half, same conventions as Table 5.